

$$\ddot{x} + 2\gamma \dot{x} + x = 0 \quad (\text{Homogeneous}) \quad *$$

3 cases

$\gamma > 1$: Over

$\gamma = 1$: Critically

$\gamma < 1$: Under

SP - units

$$m \ddot{x}_{SI} + \gamma_{SI} \dot{x}_{SI} + k x_{SI} = 0$$



$$\ddot{x} + 2\gamma \dot{x} + x = 0$$

$$\begin{aligned} \underline{T_0} &\equiv \sqrt{\frac{m}{k}} \\ \gamma &\equiv \frac{1}{2} \frac{\gamma_{SI}}{\sqrt{mk}} \end{aligned}$$

$$\underline{\gamma_{SI} < 1} : x_{SI}(t) = e^{-\omega_0 t} \left(A \cos(\omega_d t) + B \sin(\omega_d t) \right)$$

$\uparrow \quad \quad \quad \uparrow$
 $\omega_0 \quad \quad \quad \omega_d$

$$\omega_d = \omega_0 \sqrt{1 - \gamma_{SI}^2}$$

$$\omega_0 \pm i \omega_d$$

$$\underline{\gamma = 0} :$$

$$m \ddot{x} + kx = 0$$

$$\ddot{x}_{SI} = -\frac{k}{m} x_{SI}$$

ODE: SHM

$$x_{SI}(t) = A \cos(\omega t) + B \sin(\omega t)$$

$$\uparrow$$

$$\underline{\omega^2 \equiv \frac{k}{m}}$$

$$\omega^2 = \frac{1}{T_0^2}$$

$$x_{SI}(t) = A \cos(\omega t) + B \sin(\omega t)$$



$$* x(t) = A \cos\left(\frac{t}{T_0}\right) + B \sin\left(\frac{t}{T_0}\right) *$$

$$* m \ddot{x}_{SI} + \gamma_{SI} \dot{x}_{SI} + k x_{SI} = f \sin(\omega_d t) *$$

= 0 (Homogeneous)

"Driving force"

Solution: $x(t) = \underline{x_H(t)} + \underline{x_p(t)}$

$x_H(t)$: Homogeneous solⁿ

↳ solution when RHS = 0

$$m\ddot{x}_{st} + \gamma_{st}\dot{x}_{st} + kx_{st} = 0$$

$x_p(t)$: particular solⁿ

↑ guessing or by just looking at RHS and working backwards

(i) Go from st → unless,

$$T_0 \equiv \sqrt{\frac{m}{k}}$$

$$\gamma = \frac{1}{2} \frac{\gamma_{st}}{\sqrt{m k}}$$

$$\rightarrow \ddot{x} + 2\gamma\dot{x} + x = 0$$

$$\downarrow \frac{d^2 x}{d\tilde{t}^2}$$

$$\tilde{t} \equiv \frac{t}{T_0}$$

ex: $\ddot{x} + 2\gamma\dot{x} + x = C$, C : constant

$x(t) = x_p + x_H$ ← depends on γ (3-cases)

• x_H : we know this the problem just needs to tell us what γ is

$$\ddot{x} + 2\gamma\dot{x} + x = C$$

$$\begin{cases} x_p(t) = C \\ \dot{x}_p(t) = 0 \\ \ddot{x}_p(t) = 0 \end{cases}$$

$$x = x_H + C$$

$$\ddot{x} + 2\gamma\dot{x} + x - C = 0$$

$$\gamma \equiv x - C$$

$$\begin{cases} \dot{y} = \dot{x} \\ \ddot{y} = \ddot{x} \end{cases}$$

$$y = x - c$$

$$\ddot{y} + 2\gamma\dot{y} + y = 0$$

$$y = \text{Homogeneous sol}^n = x - c$$

$$x = \text{Homogeneous sol}^n + c$$

* With every second order ODE: two Initial Conditions

$$\text{Pg 52} \quad \ddot{x} + 2\gamma\dot{x} + x = 1 \quad \left| \begin{array}{l} x(0) = 0 \\ \dot{x}(0) = 0 \end{array} \right. \quad \begin{array}{l} \gamma: \text{under} \\ \gamma < 1 \end{array}$$

$$x(t) = 1 + e^{-\gamma t} (A \cos(\omega_0 t) + B \sin(\omega_0 t))$$

$$\bullet x(0) = 0$$

$$= 1 + (A) \Rightarrow \underline{A = -1}$$

$$\dot{x}(t) = -\gamma e^{-\gamma t} ((-1) \cos(\omega_0 t) + B \sin(\omega_0 t)) + e^{-\gamma t} ((-1)(-\omega_0) \sin(\omega_0 t) + B \omega_0 \cos(\omega_0 t))$$

$$\dot{x}(0) = -\gamma(-1) + B\omega_0 = 0$$

$$\underline{B = -\frac{\gamma}{\omega_0}}$$

$$x(t) = 1 - e^{-\gamma t} \left(\cos(\omega_0 t) + \frac{\gamma}{\omega_0} \sin(\omega_0 t) \right)$$

$$\omega_0 = \sqrt{1 - \gamma^2}$$

ex2: (1) Go from SE $\rightarrow$ Unless

$$\ddot{x} + 2\gamma\dot{x} + x = f(t)$$

$$f(t) = \sin(\omega t)$$

$$\ddot{x} + 2\gamma\dot{x} + x = \sin(\omega t)$$

$$x(t) = x_{\text{hom}}(t) + x_{\text{part}}(t)$$

$$\ddot{x}(t) = \ddot{x}_H(t) + \ddot{x}_p(t)$$

try:

$$(a) \cos(\omega t) = x$$

$$(b) \sin(\omega t) = x$$

$$\dot{x} = \omega \cos(\omega t)$$

$$\ddot{x} = -\omega^2 \sin(\omega t)$$

$$-\omega^2 \sin(\omega t) + 2\gamma \omega \cos(\omega t) + \sin(\omega t) = \sin(\omega t)$$

$$\text{Sol}^n: \quad c \cos(\omega t) + d \sin(\omega t) = x_p(t) \quad *$$

$$\dot{x}_p(t) = -\omega c \sin(\omega t) + d \omega \cos(\omega t)$$

$$\ddot{x}_p(t) = -\omega^2 (c \cos(\omega t) + d \sin(\omega t))$$

$$-\omega^2 (c \cos(\omega t) + d \sin(\omega t)) + 2\gamma \omega (-\omega c \sin(\omega t) + d \omega \cos(\omega t)) + c \cos(\omega t) + d \sin(\omega t) = \sin(\omega t)$$

$$\text{Collect terms: } \sin(\omega t) () + \cos(\omega t) () = 0$$

$$\sin(\omega t) (c(1-\omega^2) - 2\gamma \omega d - 1) + \cos(\omega t) (d(1-\omega^2) + 2\gamma \omega c) = 0$$

(1) just solving "by hand"

(2) Using matrices

$$\left\{ \begin{array}{l} (1) \quad c(1-\omega^2) - 2\gamma \omega d - 1 = 0 \\ \text{and} \\ d(1-\omega^2) + 2\gamma \omega c = 0 \end{array} \right\}$$

want to solve for c and d in terms of ω and γ

$$(2) \text{ Using Matrices } \begin{pmatrix} 1-\omega^2 & -2\gamma\omega \\ 2\gamma\omega & 1-\omega^2 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

$$\underbrace{\begin{pmatrix} 2\gamma\omega & (1-\omega^2) \end{pmatrix}}_W \begin{pmatrix} 1 \\ d \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$W \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$W^{-1} W \begin{pmatrix} c \\ d \end{pmatrix} = W^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} c \\ d \end{pmatrix} = W^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$W^{-1} = \frac{1}{\det(W)} \begin{pmatrix} (1-\omega^2) & 2\gamma\omega \\ -2\gamma\omega & (1-\omega^2) \end{pmatrix}$$

$$\det(W) = (1-\omega^2)^2 - (-1)(2\gamma\omega)^2$$

$$= (1-\omega^2)^2 + (2\gamma\omega)^2$$

$$W^{-1} = \frac{1}{(1-\omega^2)^2 + (2\gamma\omega)^2} \begin{pmatrix} (1-\omega^2) & 2\gamma\omega \\ -2\gamma\omega & (1-\omega^2) \end{pmatrix}$$

$$\begin{pmatrix} c \\ d \end{pmatrix} = \frac{1}{\det(W)} \begin{pmatrix} (1-\omega^2) & 2\gamma\omega \\ -2\gamma\omega & (1-\omega^2) \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \frac{1}{\det(W)} \begin{pmatrix} 1-\omega^2 \\ -2\gamma\omega \end{pmatrix}$$

$$c = \frac{1-\omega^2}{(1-\omega^2)^2 + (2\gamma\omega)^2}$$

$$d = \frac{-2\gamma\omega}{(1-\omega^2)^2 + (2\gamma\omega)^2}$$

$$X(t) = X_H(t) + c \cos(\omega t) + d \sin(\omega t)$$

* From there :

$$\left\{ \begin{array}{l} \text{(a) } \gamma: \text{ value} \\ \text{(b) I.C.s: } X(0) = ? \quad \dot{X}(0) = ? \end{array} \right\}$$

